

ARinVR: Bringing Mobile AR into VR

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ABSTRACT

Augmented Reality (AR) and Virtual Reality (VR) techniques are widely used for people to experience and interact with 3D digital content. However, the user has to choose either AR or VR separately with different devices for different interaction modalities. This work presents *ARinVR*, a novel concept that seamlessly and synchronously integrates handheld mobile AR and head-worn VR to allow users to see an augmented scenario on a phone screen and in a VR space at the same time. We developed a prototype system that enabled users to operate a smartphone and observe mobile AR scenarios in VR environment as naturally as in the real world. The system extends the AR space and occludes the physical surroundings, enabling users to stay focused on task performance and enhancing spatial perception.

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1 INTRODUCTION

Mobile phones and tablets are the most common consumer Augmented Reality (AR) devices, enabling users to observe the real world overlaid with 3D digital content in real time. The handheld mobile devices allow users to observe the AR space from different perspectives, however the small screen size greatly limits user performance and experience. On the other hand, Virtual Reality (VR) enables users to interact with digital content in a virtual environment, and current consumer VR head-mounted displays (HMDs) (e.g., the Oculus Quest) provide an immersive user experience with a wide field of view. However, users in VR are entirely separated from the real world to some extent. In our research, we explore how to combine the benefits of handheld mobile AR and head-worn

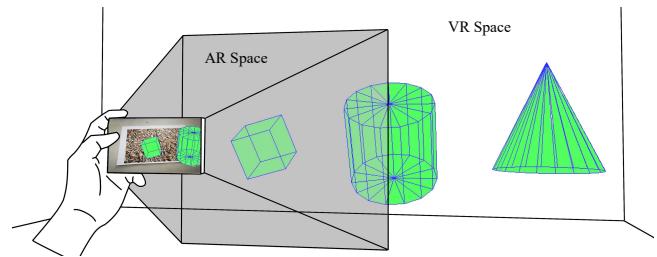


Figure 1: ARinVR is to integrate mobile AR and VR spaces to allow users to operate the content in the VR space and see its AR effect on the phone screen at the same time.

VR in a single user experience. Currently, users can only choose to experience mobile AR with smartphones and tablets or VR with head-mounted displays (HMDs) individually.

Prior research [Langner et al. 2021; Zhu and Grossman 2020] has combined smartphones with AR HMDs to extend the smartphone-based 2D content into a larger 3D AR space. Bai et al. [Bai et al. 2021] brought the full-featured smartphone into VR with controller-based tracking, screen mirroring, and virtual hand rendering. They also demonstrated the potentials of providing a smartphone camera view for physical walking and obstacle avoidance in VR [Zhang et al. 2021]. These research shows that it is technically possible to introduce smartphone-based interactions into VR.

In this work, we propose the concept of *ARinVR*, and explore how to bring mobile AR into VR so that users can observe real-time smartphone-based AR scenarios in the virtual environment as well as in the real world. As shown in Figure 1, the AR space identified by the smartphone becomes a nested subspace of the virtual world, where virtual objects are synchronously shared in the overlapped region. Therefore, the AR scenario presented on the 2D phone screen can be viewed in the extended VR space, and users can focus on smartphone-based interactions. Conversely, the content can be designed and implemented in the VR space, and moved to the AR subspace for users to see the AR effect on the smartphone. In this paper, we present the design and implementation of a prototype system and user experience evaluation.

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2 PROTOTYPE DESIGN

The system brings mobile AR scenarios into VR and enables users to see and touch the smartphone screen with their real hands. The system mainly consists of three components, 1) smartphone virtualization, 2) content synchronization, and 3) hand rendering. The system uses an Oculus Quest2 VR HMD with an Intel SR305 depth sensor attached and an Android smartphone for tracking and viewing AR content.

(1) Smartphone virtualization

The mobile AR runs on a smartphone and uses an image marker for 3D tracking. The system attaches a VR controller to the smartphone to provide large-scale 3D tracking (see Figure 2), so that the AR and VR spaces are combined. The real phone screen is mirrored onto the front panel of a virtual phone model as a high-definition live texture with low latency. When the real phone moves around the marker to view AR content, the virtual phone model moves correspondingly and presents a clear AR view in the VR space.

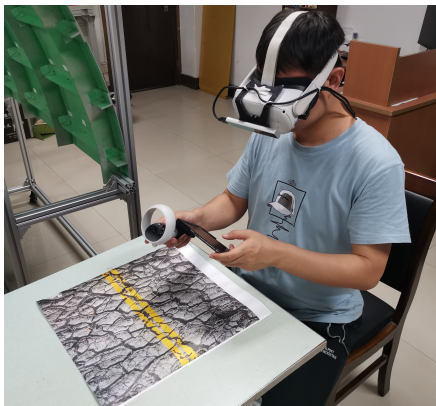


Figure 2: The physical setup of the integrated system.

(2) Content synchronization

Since the AR and VR applications are running on different devices, the system sets up a TCP/IP communication channel to synchronize the contents between VR and AR spaces. Commands and messages (e.g., touchpoints, object positions) are synchronously sent to each other when an input is detected. When moving an object to the AR subspace from beyond the region, users can see synchronous AR presentation on the phone screen and VR presentation from the HMD.

(3) Hand rendering

The RGB-Depth camera is horizontally installed in the front panel of the VR HMD to capture the user's hands as a point cloud set. A color-based filter is applied to reserve only the skin-colored region, being rendered in the VR space to enable users to see their own hands with a better embodiment. This is calibrated with the HMD to align the rendered virtual hands with the user's real hands.

3 USER EXPERIENCE

We present a typical *ARinVR* application to demonstrate the combination of mobile AR and VR, as shown in Figure 3). The initial

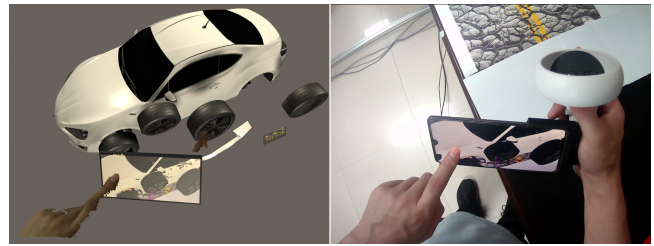


Figure 3: A typical *ARinVR* scenario enables users to see the AR effect while operating in VR environment. It also separates the visualization and interaction to different devices.

AR scenario is to assemble a complex virtual model with parts distributed in a large AR space. The user holds the phone to find and move these parts with raycast-based selection. Limited by the screen size, users need to spend extra time searching for targets beyond the field of view and positioning the phone to observe the whole appearance of large objects. This system extends the AR space for users to see more virtual content beyond the phone screen with the large field of view VR HMD. The efficiency of object finding and spatial perception could be greatly improved.

The system provides a solution to separate the integrated interaction and visualization in mobile AR. Users can see the augmented virtual content from the VR space rather than on the phone screen and only use the phone for interactions. Thus they can focus on the task performance in the 3D VR space with quick and precise input with smartphone-based interactions. The phone screen is only used to observe the AR effect until a specific operation is completed.

4 CONCLUSION AND FUTURE WORK

This work proposes to bring mobile AR into VR by breaking the boundary between these spaces. The system enables users to observe and interact with AR content in a larger 3D VR space, and helps users stay focused on the task performance without being distracted. Users can also benefit from separated visualization and interaction from different devices to enhance spatial perception.

The system brings mobile AR into the virtual space together with smartphone-based interactions, but the controllers and hand gestures can also provide rich input. In the future, we plan to investigate more potential applications with this system and conduct an exhaustive user study to evaluate its usability by comparing it with current mobile AR approaches.

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